

SARTHAK PANDEY

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EDUCATION

Scaler School Of Technology

4 year undergraduate program

Expected 2027

CGR: 8.11

BITS Pilani

BE / B.Tech / BS

Expected 2027

CGPA: 9.0

St. Thomas' School

Class XII (ISC)

2021

90.2%

St. Thomas' School

Class X (ICSE)

2019

94.6%

EXPERIENCE

Scaler

SWE Intern

Apr 2025 – March 2026

India

- Architected and deployed a production-grade campus management system (Next.js, TypeScript) used by 2,000+ users, gaining deep experience in concurrent-safe workflows and low-latency API serving.
- Engineered internal workflow tools and debugged real-world production issues across backend pipelines, improving operational efficiency and system reliability.
- Designed multi-role systems with efficient state management, ensuring smooth coordination across complex, multi-step workflows (highly relevant for stateful agentic pipelines).

TradeIndia

Product Development Intern

Jan 2025 – Mar 2025

India

- Engineered and integrated conversational AI-driven voice systems to automate lead qualification, directly aligning with autonomous user interaction workflows.
- Optimized LLM context management and backend interactions to reduce latency, improving the flow of real-time voice interactions.
- Designed strategic “exit-intent” features and automated inquiry popups, driving increased engagement and boosting lead conversion rates.

PROJECTS

MrBully: Autonomous Accountability Agent

Kotlin, Groq API, Room DB

- Built an Android accountability application utilizing Accessibility Services to monitor foreground usage and trigger real-time, autonomous psychological interventions.
- Integrated the Groq API for rapid LLM inference, strictly parsing AI outputs to determine phrase-based unlocks and stop app distractions.
- Engineered a secure local architecture using DataStore with AES/GCM encryption for sensitive LLM API key management.

FinTracker: AI-Enhanced Financial Platform

Next.js, JavaScript, LLM APIs

- Developed a personal finance management application featuring an added layer of AI-driven insights to autonomously track and categorize expenses.
- Built API-driven backend integrations to process user data through LLMs, generating personalized budgeting feedback.

StoryTeller: Agentic Story Generation

Next.js, TypeScript, React

- Built an interactive storytelling platform powered by AI, demonstrating multi-step reasoning where AI generates continuing narratives based on user prompts.
- Implemented persistent state management and structured API data handling to maintain context windows across continuous user-AI interactions.

SSTBORROWING: Production Booking System

TypeScript, Next.js, APIs

- Deployed a unified campus booking system handling multi-role workflows, automated penalty management, and QR-based check-ins for 2,000+ users.
- Implemented concurrent-safe backend logic and scalable UI systems for real-time resource allocation.

ACHIEVEMENTS

- Completed Hacktoberfest 2024
- Member – ZOPE Foundation

TECHNICAL SKILLS

- **LLM & AI Engineering:** LLM APIs (Groq, OpenAI, Claude, Gemini), Prompt Engineering (CoT, ReAct, RAG prompting), Decoding Strategies (Top-p, Top-k, Temperature), LLM-as-Judge Evaluation, Agentic Pipelines, Function Calling, Context & Token Management.
- **RAG & Retrieval:** Retrieval-Augmented Generation (RAG), Chunking Strategies (fixed-size, semantic, recursive), Vector Embeddings, Semantic Search, HyDE, Bi-encoder Retrieval, Cross-encoder Reranking, ANN / HNSW indexing.
- **Vector DBs & Embeddings:** Pinecone, Chroma, Weaviate (conceptual); Cosine Similarity, SBERT, Multimodal Embeddings (CLIP), Embedding fine-tuning, Hard Negatives, Retrieval Metrics (Precision@K, NDCG@K, MRR@K).
- **AI Agents:** Agent Loop (Perceive → Reason → Act → Observe), Tool Use, Stateful Multi-step Workflows, Automated vs. Agentic Pipelines, Guardrails, Human-in-the-loop.
- **Evaluation & Observability:** BLEU, ROUGE, BERTScore, LLM-as-Judge, Offline + Online Evals, Experiment Tracking (LangSmith, Langfuse), Latency & Cost Monitoring.
- **Security:** Prompt Injection (direct & indirect), OWASP GenAI Top 10, Input/Output Sanitization, Least Privilege for Agents, Llama Guard, Sensitive Data Handling.
- **Backend & APIs:** Python, Node.js, REST APIs, Async & Concurrent Systems, Firebase, MongoDB, MySQL, Room DB.
- **Frontend:** TypeScript, Next.js, React.js, Jetpack Compose, Flutter, HTML, CSS.
- **Languages:** Python, TypeScript, JavaScript, Kotlin, Java, C++, Golang, Dart.
- **Tools:** Git, GitHub, Docker, LangChain (conceptual), LlamaIndex (conceptual), Chrome DevTools.